Model Justification

Multinomial Naive Bayes was selected as a baseline model because it is computationally efficient and commonly used for text classification. It estimates the probability of a sentiment class based on the words appearing in a review. Although it assumes that features are conditionally independent given the class, which is a simplification of natural language, it provides a useful benchmark for evaluating more advanced classifiers.

Logistic Regression was selected because it is an effective linear classification algorithm for high-dimensional text data. It learns the relationship between textual features and sentiment labels and estimates the probability that a review belongs to a particular class. It is also relatively efficient to train and provides a strong baseline for binary sentiment classification.

Linear Support Vector Machine (SVM) was included because it is well suited to high-dimensional, sparse feature representations such as TF-IDF. It identifies a decision boundary that separates positive and negative reviews while aiming to maximize the margin between the two classes. This makes it a strong candidate for text classification and allows its performance to be compared with Logistic Regression.

Model Performance

	Logistic Regression	Linear SVM	Naive Bayes
Accuracy	*0.8990*	*0.8944*	*0.8692*
Precision	*0.8991*	*0.8944*	*0.8694*
Recall	*0.8990*	*0.8944*	*0.8692*
F1-score	*0.8990*	*0.8944*	*0.8692*

Table 1 - Model Performance Comparison

Based on these results, Logistic Regression is the recommended model for this experiment, as it achieved the highest Accuracy as well as Precision, Recall and F1-score.

Linear SVM is a close second and could also be a suitable choice. Naive Bayes performed less well but provides a useful baseline for demonstrating how different classification algorithms handle the same text features.

Confusion Matrices

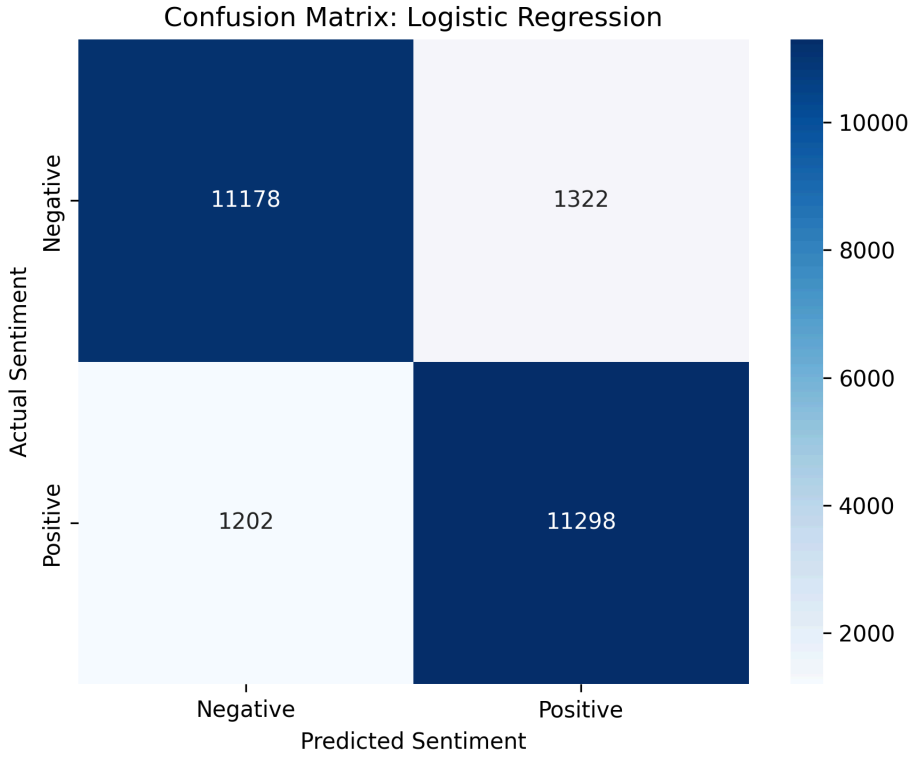


Figure 1 - Confusion Matrix of Logistic Regression

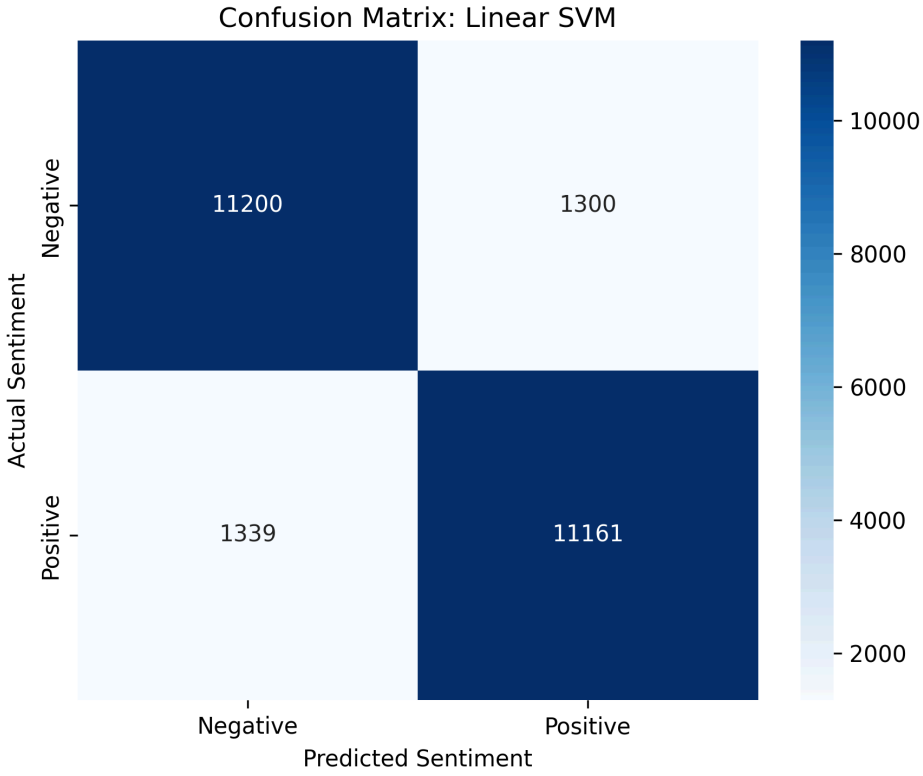


Figure 2 - Confusion Matrix of Linear SVM

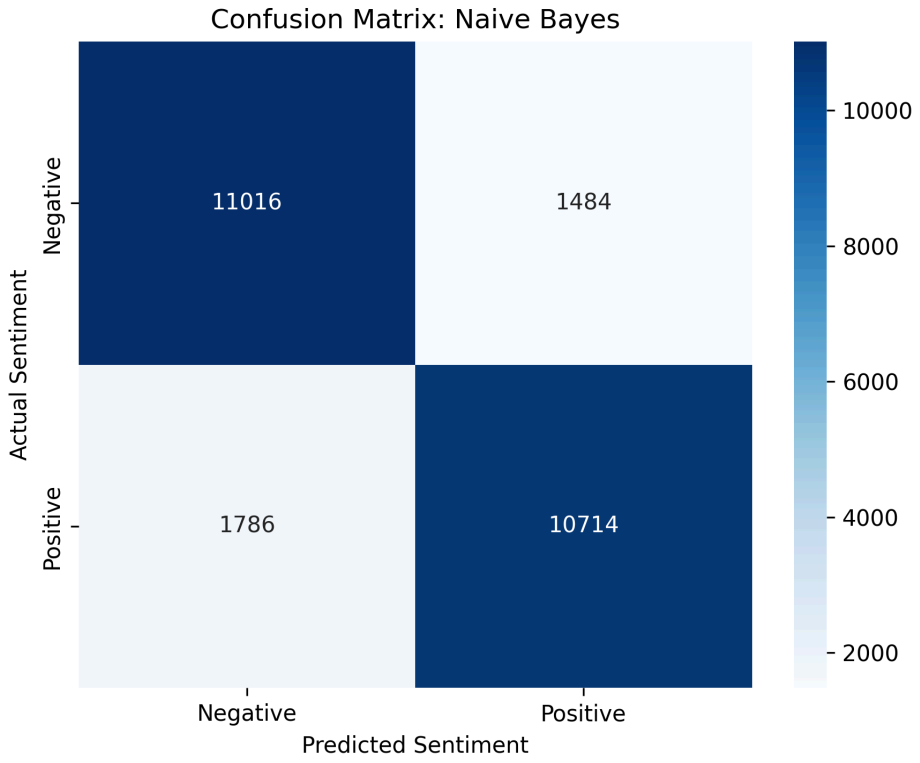


Figure 3 - Confusion Matrix of Naive Bayes

	Logistic Regression	Linear SVM	Naive Bayes
Correct Predictions	22,476	22,361	21,730
Incorrect Predictions	2,525	2,639	3,270
Negative reviews misclassified as positive	1,322	1,300	1,484
Positive reviews misclassified as negative	1,202	1,339	1,786
Accuracy	86.92%	89.44%	89.90%

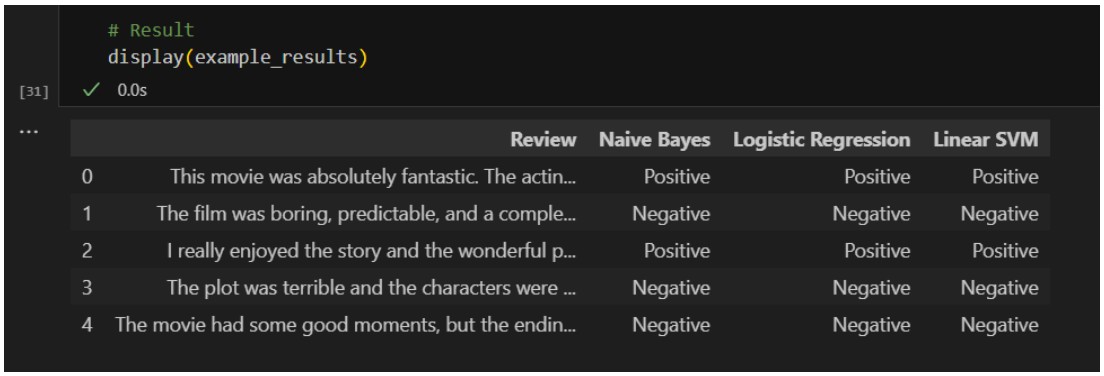
Table 2 - Model Confusion Matrix Comparison

Naive Bayes correctly classified 11,016 negative reviews and 10,714 positive reviews. However, it misclassified 1,484 negative reviews as positive and 1,786 positive reviews as negative. These results correspond to an accuracy of 86.92%. The larger number of positive reviews incorrectly classified as negative suggests that Naive Bayes was less effective at identifying positive sentiment in this experiment.

Linear SVM achieved better results, correctly classifying 11,200 negative reviews and 11,161 positive reviews. It produced 2,639 incorrect predictions, resulting in an accuracy of 89.44%. Its misclassification counts were relatively balanced, with 1,300 negative reviews predicted as positive and 1,339 positive reviews predicted as negative. This demonstrates that Linear SVM performed consistently across both sentiment classes.

Logistic Regression achieved the highest overall accuracy of 89.90%, correctly classifying 22,476 out of 25,000 reviews. It misclassified 1,322 negative reviews as positive and 1,202 positive reviews as negative, resulting in 2,524 errors. Although Linear SVM made slightly fewer errors when identifying negative reviews, Logistic Regression made fewer errors overall, particularly when identifying positive reviews.

Testing Result



```
# Result
display(example_results)
```

[31] ✓ 0.0s

...

	Review	Naive Bayes	Logistic Regression	Linear SVM
0	This movie was absolutely fantastic. The actin...	Positive	Positive	Positive
1	The film was boring, predictable, and a comple...	Negative	Negative	Negative
2	I really enjoyed the story and the wonderful p...	Positive	Positive	Positive
3	The plot was terrible and the characters were ...	Negative	Negative	Negative
4	The movie had some good moments, but the endin...	Negative	Negative	Negative

Figure 4 - Testing Result for 5 custom reviews

5 custom reviews used:

1. "This movie was absolutely fantastic. The acting was brilliant!"
2. "The film was boring, predictable, and a complete waste of time.",
3. "I really enjoyed the story and the wonderful performances.",
4. "The plot was terrible and the characters were poorly written.",
5. "The movie had some good moments, but the ending was disappointing."